

FA25ProgressReport_Lucas

2025/11/24

Interlocking Topology

Coded an automated process that used a traversal method from top to bottom to classify and label segments, calculates the sum of the segment lengths in each image, and uses this to compute three Geometric Descriptors.

Geometric Classification

Segments are classified into four mechanical categories based on their orientation angle θ relative to the horizontal axis:

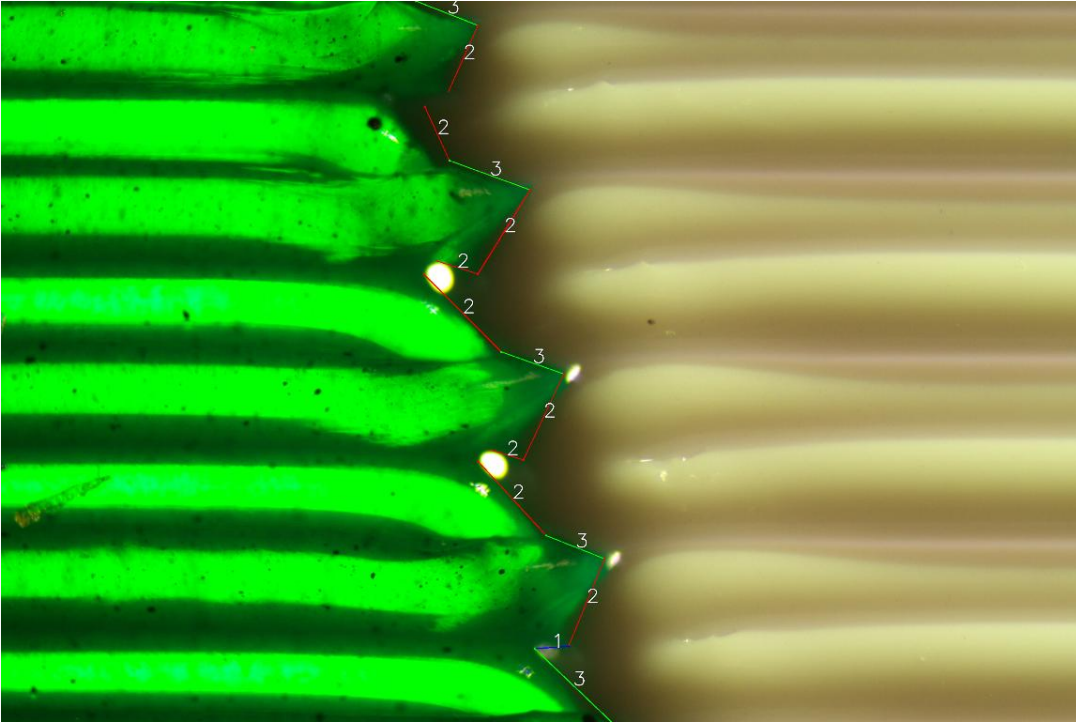
Category	Classification	Angle Range (θ)	Function / Surface Type
1	Horizontal	$ \theta \leq 10^\circ$	Primary Interlock Surface
2	Closing	$ \theta > 45^\circ$	Adhesion Interface (Closing)
3	Opening Upper	$10^\circ < \theta \leq 45^\circ$	Adhesion Interface (Opening)
4	Opening Lower	$-45^\circ \leq \theta < -10^\circ$	Adhesion Interface (Opening)

Quantitative Descriptors

Calculates key metrics to quantify mechanical strength potential:

SAEF	Surface Area Enhancement Factor	The ratio of total actual interface length to the projected seam length ($L_{total}/L_{projected}$).
MID	Mechanical Interlocking Depth	The maximum amplitude of the interface in the interlocking direction (X-axis span).
VIR	Volumetric Interlock Ratio	Normalized ratio representing the effective geometric overlap area relative to the bounding box defined by MID and projected length.

Interlocking Topology



Found 15 valid segments.

--- Segment Statistics (mm) ---

Total CAT_HORIZONTAL Length: 0.1093

Total CAT_CLOSING Length: 2.2933

Total CAT_OPENING_UPPER Length: 1.2516

Total CAT_OPENING_LOWER Length: 0.0000

--- Geometric Descriptors ---

SAEF: 1.5709

MID: 0.6307 mm

VIR: 0.2299

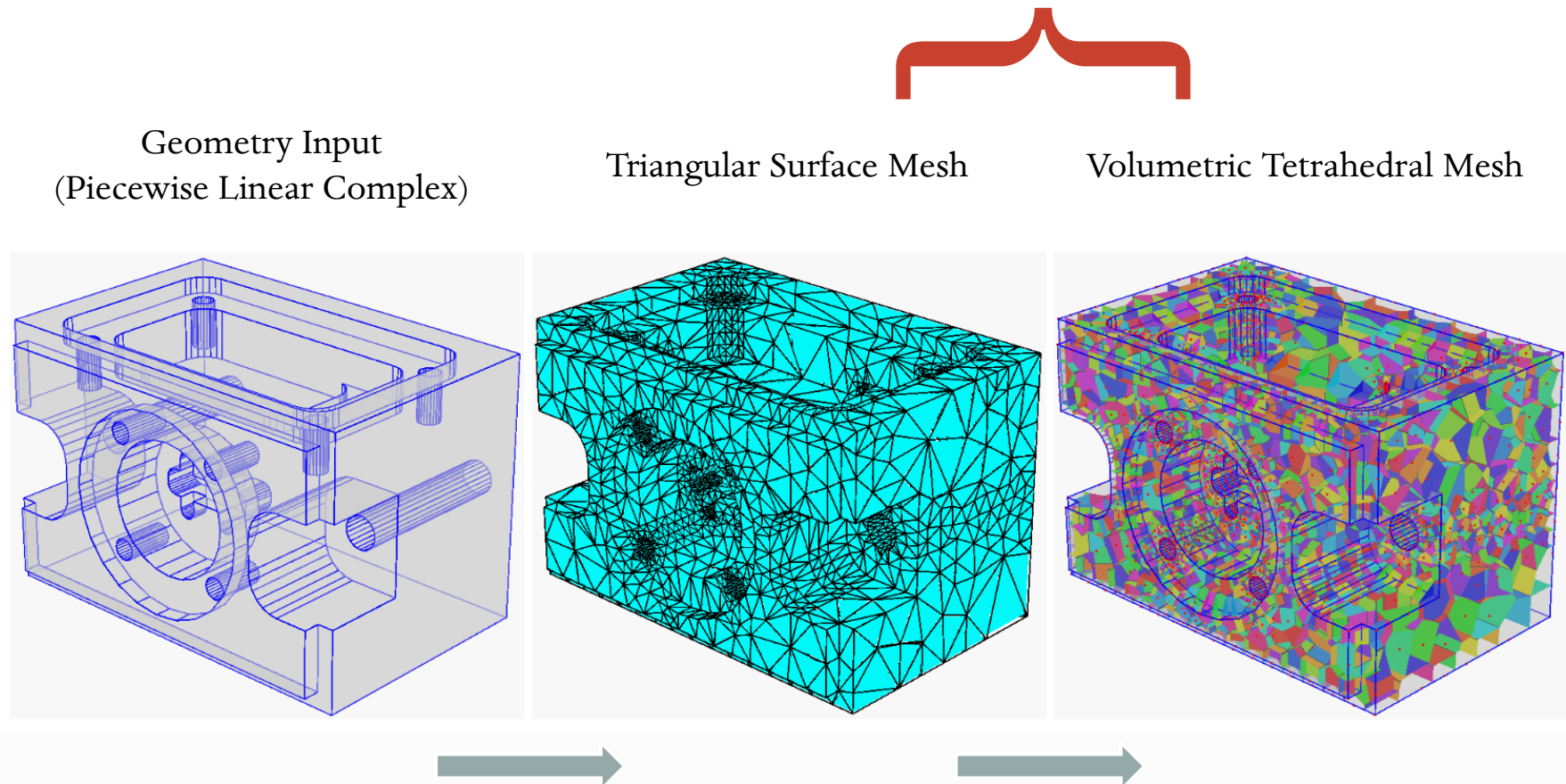
- ♦ Categorized and Labeled Segments

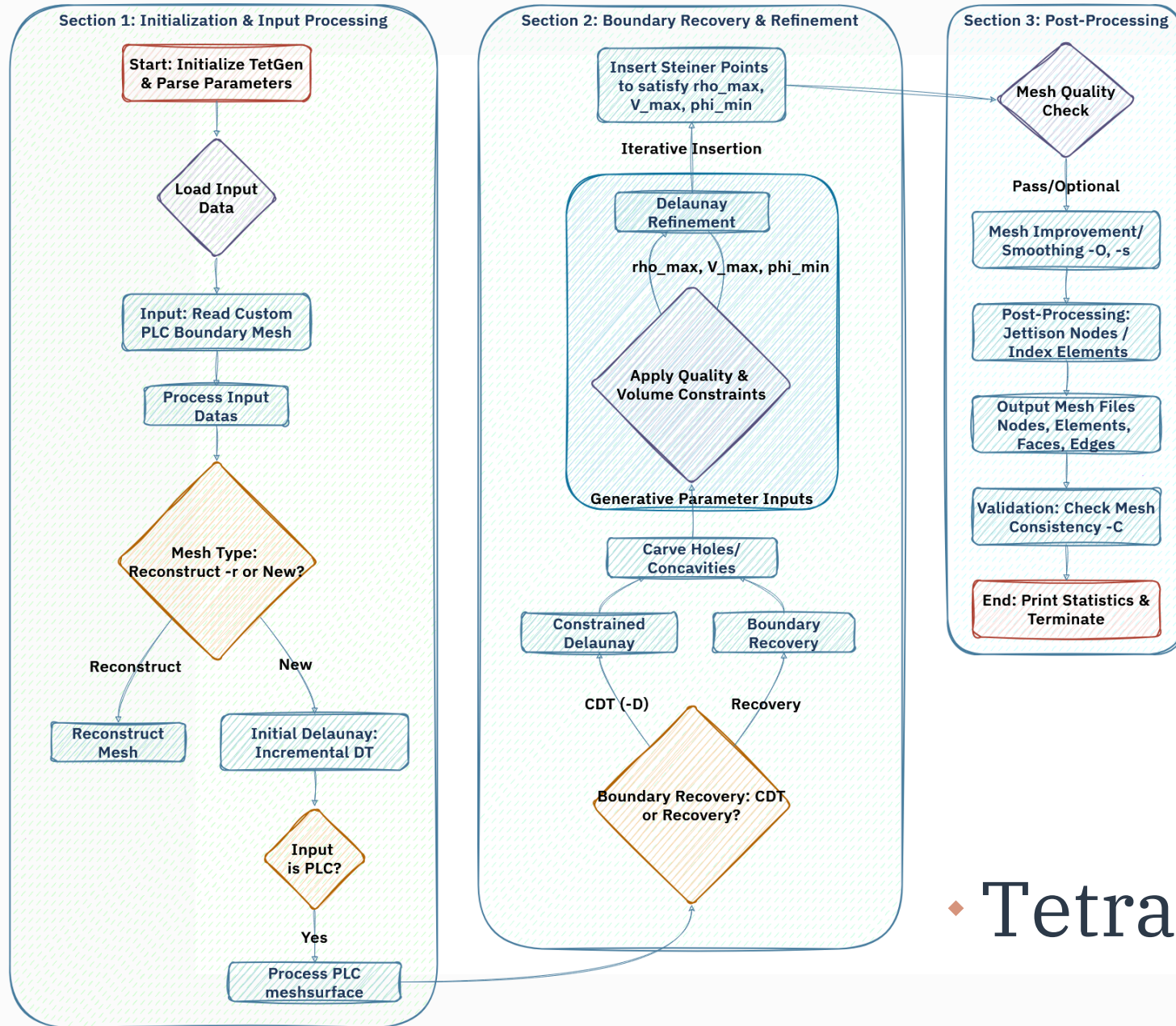
- ♦ Statistics Output

Learning FEA Workflow

— Mixed mesh model Simulation and Optimisation

- ♦ Mesh generation and combination



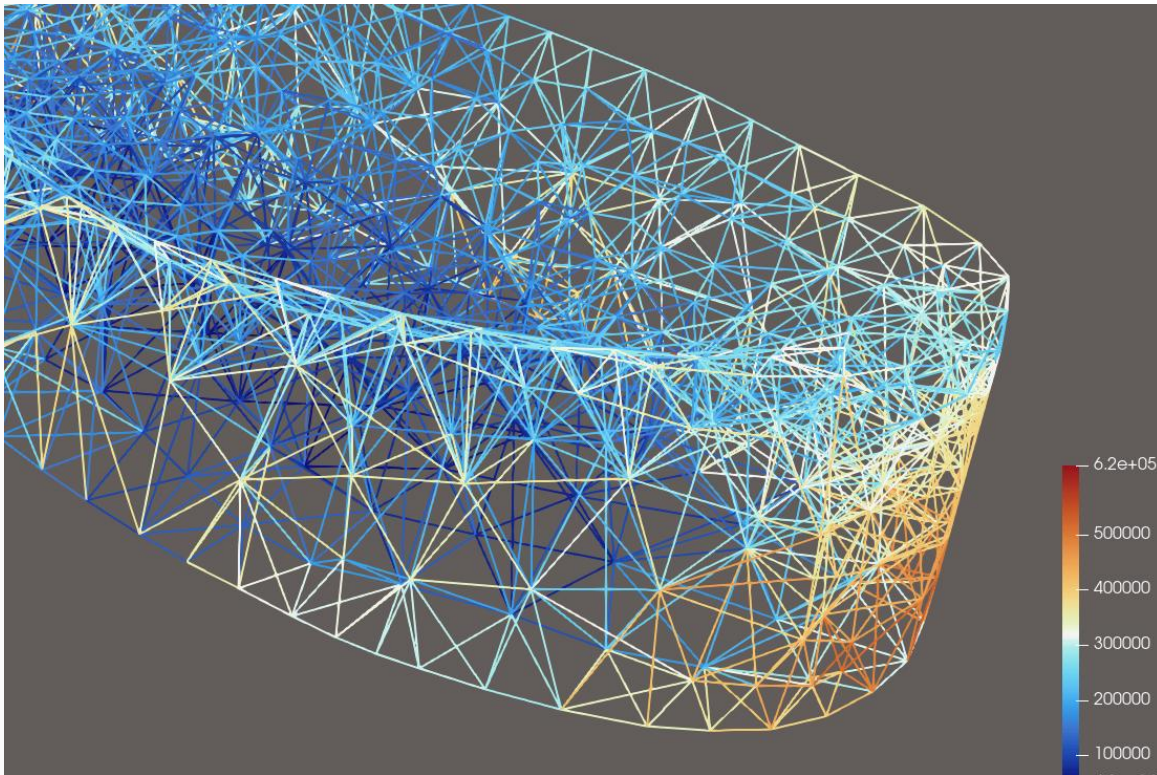


◆ Tetrahedral Mesh Workflow

Pyvista – Tetgen – gmsh – FEniCS Toolchain

Result Visualisation in Paraview

- Von Mises stress plot



- Displacement (Z) plot

